EXP :- 3

i) Force B/w Two Electric Charge

Charge 1 = 3×10^{-6}

Charge 2 = 5×10^{-6}

r = 3 cm 0.03 m

Current 1 = 2 A

Current 2 = 3 A

L = 1 mm = 0.001 m

D = 10 m

Formula :- $\frac{1}{4\pi\varepsilon_0} \cdot \frac{q_1 \cdot q_2}{r^2}$

given, $q_1 = 3\times10^{-6} C$

$q_2 = 5\times10^{-6} C$

$r = 0.03 m$

$\varepsilon_0 = 8.854\times10^{-12} F/m$

Coulomb const $K = \frac{1}{4\pi\varepsilon_0} = 8.99\times10^{9} N m^2/C^2$

Calculation :-

$$F = (8.99\times10^{9}) \times \frac{(3\times10^{-6})(5\times10^{-6})}{(0.03)^2}$$

$$F = (8.99\times10^{9}) \times \frac{15\times10^{-12}}{9\times10^{-4}}$$

$$F = (8.99\times10^{9}) \times (1.67\times10^{-8})$$

$F = 150 N$ //

$\therefore 149.8 N \approx 150 N$

ii) Fource b/w two Parallel condition

Formula:- $$F = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

$\mu_0 = 4\pi \times 10^{-7}$ H/m

$$F = \frac{(4\pi \times 10^{-7})(2)(3)(0.01)}{2\pi(10)} \Rightarrow F = \frac{24\pi \times 10^{-10}}{20\pi}$$

$F = 1.2 \times 10^{-10}$ N//